

What Should SRE Watch?

Calibration-Gated Decisions and the Missing Golden Signals for GPU Clusters

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scan for live demo

Queue wait is the user-visible latency on a shared GPU cluster. GPU utilization does not see it. We measure Queue-of-Work Reliability (QWR) — a calibrated $P(\text{wait} > \text{SLO})$ — and gate actions on its ECE. The live production model is at ECE 0.068: both models land in T2 Warn.

THE FOUR TRAVEL POORLY

Classical signals

Google SRE’s four — latency, traffic, errors, saturation — assume a stateless request/response service. On shared GPU clusters:

Latency has token, batch, epoch, *and queue* meanings. Submit-to-start often dominates. HTTP p99 never sees it.

Traffic as QPS is undefined for training. Tokens/s and active jobs are different loads.

Errors as HTTP 5xx miss silent quality failure (200 + a bad output) and OOM / throttle restarts.

Saturation as `nvidia-smi` utilization is time-busy, not useful compute. Memory-bound kernels, thermal trips, and NCCL stragglers can all look fully utilized.

Mechanistic failure modes. The Slurm DCGM slice is idle T4 queue-wait (utilization = 0); a util-goodput correlation is undefined on that trace.

A PAGING-ROTATION MINIMUM

Five candidates

QWR latency / queue

At submit: $\hat{p} = P(\text{wait} > \text{SLO})$. Rolling ECE of $\hat{p}$. T3 Suggest < 0.05; T4 Gate < 0.03.

Measured — held-out Slurm + live VGAC

GCE saturation / compute

Util dispersion, thermal spread, power-brake rate. Utilization is one feature, not the signal.

Defined — needs non-idle DCGM

HPH saturation / HBM

Headroom remaining plus its derivative, catching OOM cliffs before they page.

Defined — same idle-trace limit

SAI latency / collectives

P_{90}/P_{50} step time across workers.

Candidate — no per-worker step times

OQC errors / quality

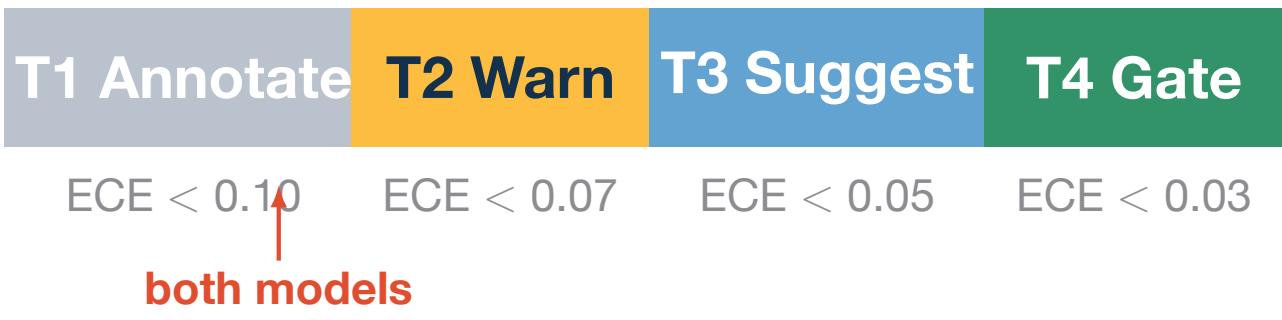
ECE of *model outputs* vs a shadow label stream. Not the queue-delay classifier.

Candidate — not measured here

THE TITLE CLAIM

Both models: T2 Warn

Actions are not “the model said 70%.” They are gated on whether that 70% is calibrated. The tier ladder from `tier_qualification.png`:



Live VGAC

v4.0-richfeatures-lr

ECE 0.068 AUROC 0.766 $n = 5,184$

T2 Warn (fails 0.05 Suggest and 0.03 Gate)

Held-out Slurm

public artifact, 444/111

GB AUROC 0.863 (CI 0.71–0.96)

ECE 0.056 T2 Warn

~8 positives in test set 0 true alerts at $p \geq 0.7$

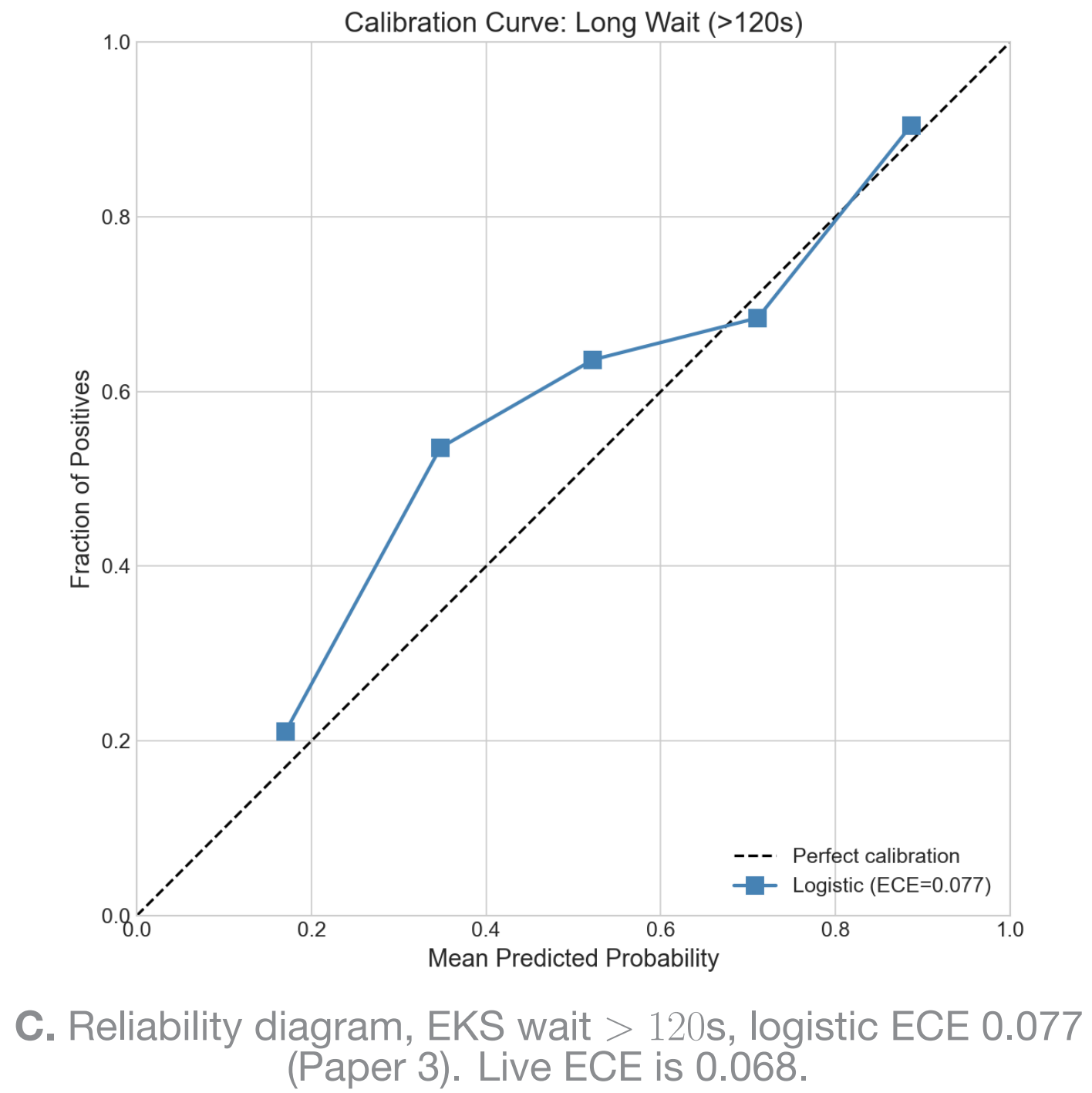
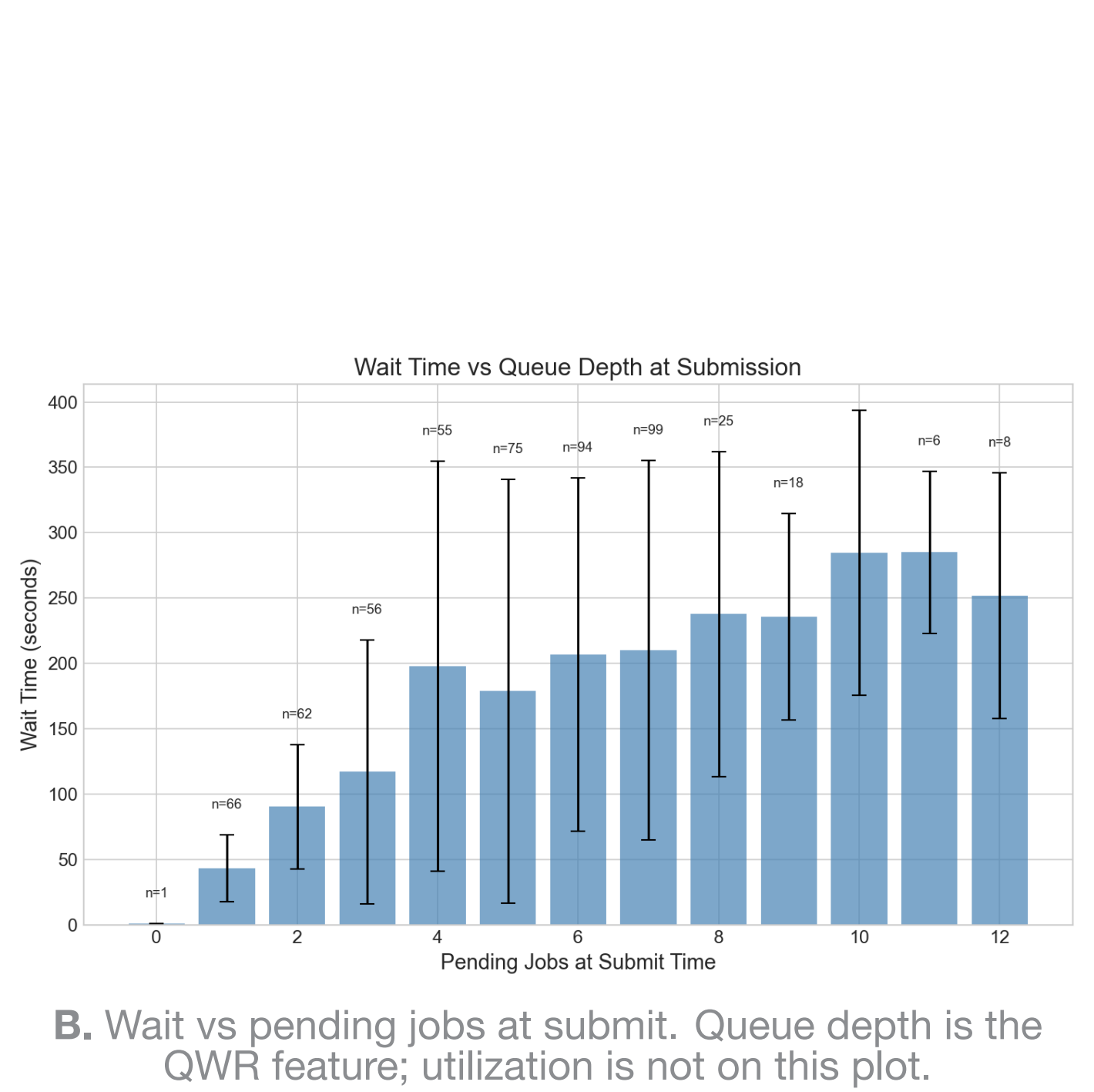
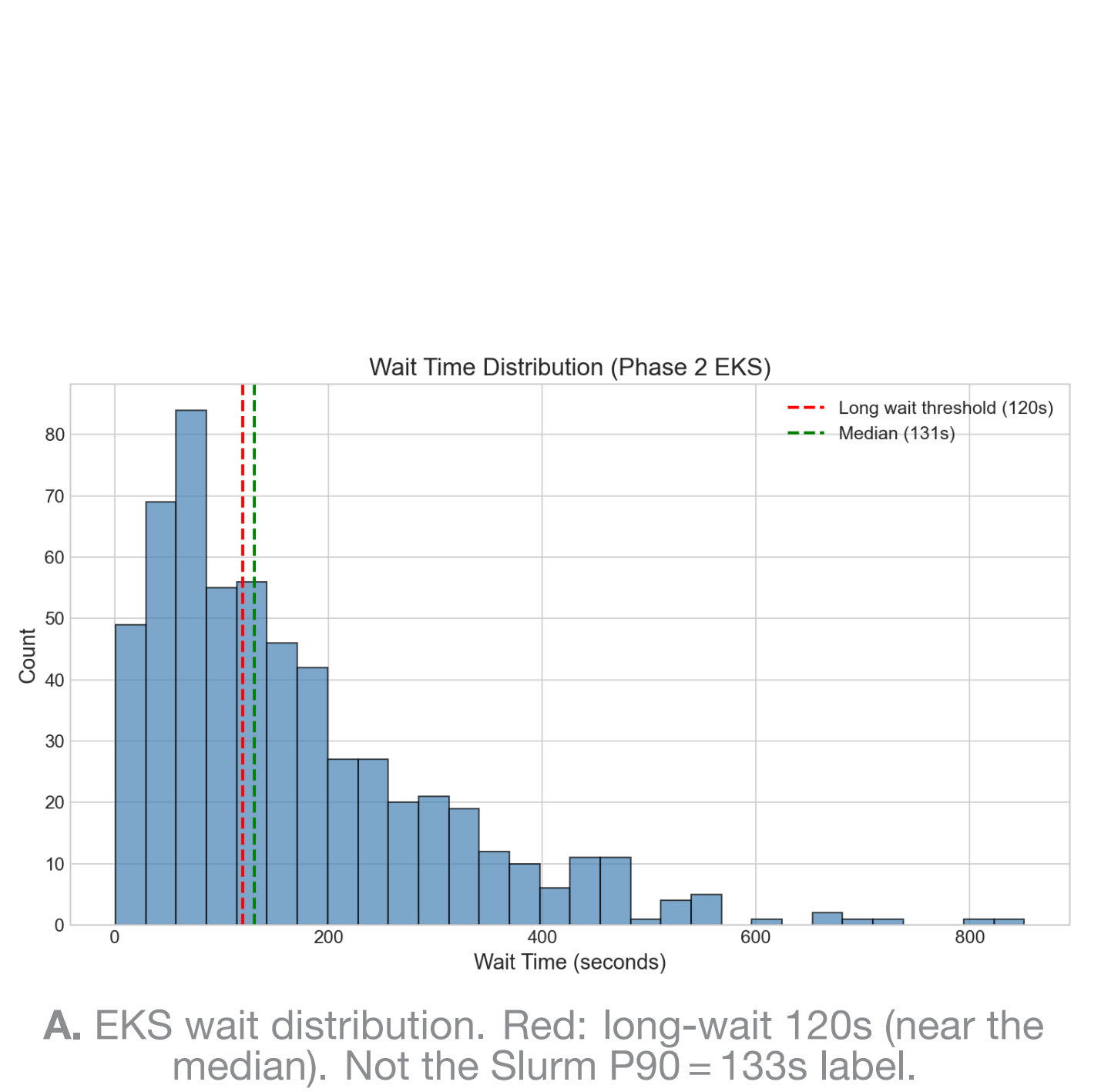
Labels differ between clusters

- EKS: wait > 120s (near the median).
- Slurm: wait > P90 = 133s (~10% base rate). Do not pool AUROC or ECE across those labels.

Figure C is Paper 3 (650 EKS jobs, ECE 0.077) — not the live 0.068 card.

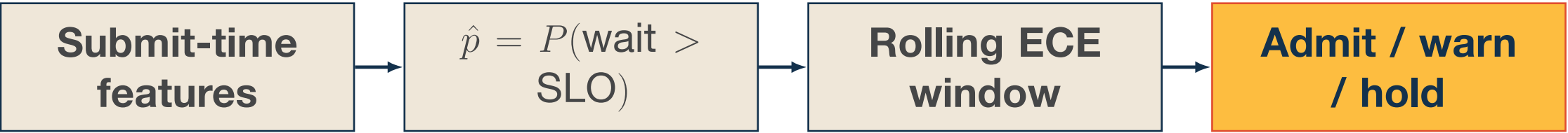
MEASURED EVIDENCE

Queue wait is the measured story



HOW AN SRE WOULD USE THIS

From $\hat{p}$ to a decision



VGAC at [demo.vgac.cloud](#) runs this loop on a t4g.small Graviton host. Predictions=QWR. Calibration=the gate (currently T2 Warn). GPUs=GCE/HPH candidates. HPC=SAI candidate.

SCOPE AND LIMITATIONS

Limits

- One signal measured (QWR). GCE, HPH, SAI, OQC are candidates only.
- Small test set: ~8 positives in 111 Slurm test jobs.
- DCGM columns idle on T4 queue-wait trace; GCE/HPH cannot be computed from it.
- EKS and Slurm use different “long wait” labels; results are not pooled.